

Architecting Solutions on AWS

Module 1  
Designing a serverless web backend on AWS

Module 2  
Designing a serverless data analytics solution on AWS

Module 3  
Designing a hybrid solution for container based workloads on AWS

Module 4  
Designing a solution following account governance and management best practices

Customer #4 - A Fast Growing Start-up Company

Week 4 Introduction  
Video • 1 min

Customer #4: Use Case and Requirements  
Video • 5 min

Customer #4: Requirements Breakdown  
Video • 2 min

Architecting the Solution

Why Multi-Account Strategies?  
Video • 3 min

Multi-Account Strategies  
Reading • 25 min

IAM Roles - The AWS Authentication Core  
Mechanism  
Video • 6 min

IAM Roles, Trust Relationships, and Permissions  
Reading • 25 min

Organizing Your Accounts Together Using OUs (Organizational Units)  
Video • 10 min

Moving from One Account to Another  
Video • 6 min

AWS IAM Identity Center  
Reading • 25 min

Reflecting on AWS Multi-Account Architecture Decisions  
Dialogue • 15 min

Choosing Services for Logging  
Video • 7 min

Logging Services  
Reading • 25 min

Automating Account Provisioning and Maintenance  
Video • 5 min

Automatic Account Provisioning  
Reading • 25 min

Multi-Account Governance Strategy  
Consultation  
Role Play • 15 min

Presenting the Solution to the Customer & Next Steps

Customer #4: Solution Overview  
Video • 2 min

Week Wrap-Up: Taking this Architecture to the Next Level  
Video • 6 min

Multi-Account Best Practices  
Reading • 25 min

Capstone Project  
Graded Assignment • 2h

Course Closing  
Video • 2 min

Final Assessment  
Graded Assignment • 1h 10m

Post-Course Survey  
Reading • 1 min

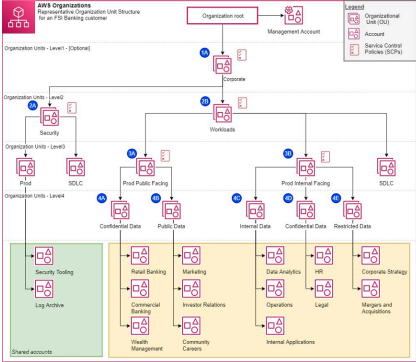
Post-Course Survey  
Ungraded Plugin • 15 min

Multi-Account Best Practices

Implementing multi-account best practices is an ongoing effort. The following links and references provide resources so that you can learn more about the subject. Remember that the goal of this course is not to dive deep into each approach. Instead, the goal of this course is to maintain knowledge as a Solutions Architect - Associate level by highlighting the services that are used for the most common approaches.

Multi-account environments

The AWS for Industries Blog includes a blog post called [Best Practices for AWS Organizations Service Control Policies in a Multi-Account Environment](#). This blog post talks about core concepts of AWS Organizations and service control policies (SCPs). It also provides suggestions about using different organizational unit (OU) structures for different use cases (such as a corporate OU, a production public-facing OU, a production internal-facing OU, and a security OU). At the end, you might have a complex structure that looks like the following diagram:



[Organizing Your AWS Environments Using Multiple Accounts](#) is an AWS Whitepaper that was published on July 26, 2022. It talks about the AWS Well-Architected Framework and much of the content that you learned about in this course. It's a good read!

The AWS Organizations website also has another reference for further reading, called [Establishing your best practice AWS environment](#).

Tag policies and SCPs

With Organizations, you can set tag policies, in addition to defining SCPs. You can use tag policies to maintain standardized tags for AWS resources that are used with Organizations accounts. For example, the following example tag policy defines a tag with a key of Environment and a value of Production, and this tag is enforced for Amazon Elastic Compute Cloud (Amazon EC2) instances.

```
{
  "tags": {
    "Environment": {
      "tag_key": [
        "@@assign": "Environment"
      ],
      "tag_value": {
        "@@assign": [
          "Production"
        ]
      },
      "enforced_for": {
        "@@assign": [
          "ec2:instance"
        ]
      }
    }
  }
}
```

This tag policy prevents users from changing this tag on existing Amazon EC2 instances. However, it doesn't prevent a user from launching new instances with non-compliant tags, or no tags. This tag policy might be adequate if you use infrastructure as code (IaC) to provision environments. With IaC, environments will be created from a CloudFormation template, and you can thus embed the tags in the template.

However, if you want to prevent the creation of new AWS resources that aren't tagged, you need to use SCPs. You could use the following example SCP to make sure that the AWS resources are created only if a certain tag is present. This SCP requires specific tags on specified created resources, and it uses explicit deny statements.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyCreateSecretWithNoProjectTag",
      "Effect": "Deny",
      "Action": "secretsmanager:CreateSecret",
      "Resource": "*",
      "Condition": {
        "Null": {
          "aws:RequestTag/Project": "true"
        }
      }
    },
    {
      "Sid": "DenyRunInstanceWithNoProjectTag",
      "Effect": "Deny",
      "Action": "ec2:RunInstances",
      "Resource": [
        "arn:aws:ec2:*:*:instance/*",
        "arn:aws:ec2:*:*:volume/*"
      ],
      "Condition": {
        "Null": {
          "aws:RequestTag/Project": "true"
        }
      }
    },
    {
      "Sid": "DenyCreateSecretWithNoCostCenterTag",
      "Effect": "Deny",
      "Action": "secretsmanager:CreateSecret",
      "Resource": "*",
      "Condition": {
        "Null": {
          "aws:RequestTag/CostCenter": "true"
        }
      }
    },
    {
      "Sid": "DenyRunInstanceWithNoCostCenterTag",
      "Effect": "Deny",
      "Action": "ec2:RunInstances",
      "Resource": [
        "arn:aws:ec2:*:*:instance/*",
        "arn:aws:ec2:*:*:volume/*"
      ],
      "Condition": {
        "Null": {
          "aws:RequestTag/CostCenter": "true"
        }
      }
    }
  ]
}
```

Do you see how SCPs and tag policies can—and should—be used together? For more information, see [Require a tag on specified created resources](#) in the AWS Organizations User Guide.

Mark as completed

coach

Hi, Venkatesh!

How can I help?

Explain this topic in simple terms

Give me a summary

Give me practice questions

Give me real-life examples

Ask me anything

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